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B. Tech. III - Sem. (Main) Exam., (Academic Session 2021- 2022)
Computer Science and Engineering
3CS3 – 04 Digital Electronics
Common with CSE/IT/AI/DS/ MLC/AI&DS/AI&ML/CSE (DS)/CSE (AI &ML)/CSE (AI)

Time: 2½ Hours

Maximum Marks: 120
Min. Passing Marks:

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 6 × 3 marks = 18 marks. Candidates have to answer **six** questions out of **ten***

Part – B: *Analytical/Problem solving questions 3 × 10 marks = 30 marks. Candidates have to answer **three** questions out of **seven**.*

Part – C: *Descriptive/Analytical/Problem Solving questions 3 ×24 marks = 72 marks. Candidates have to answer **three** questions out of **five**.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL

2. NIL

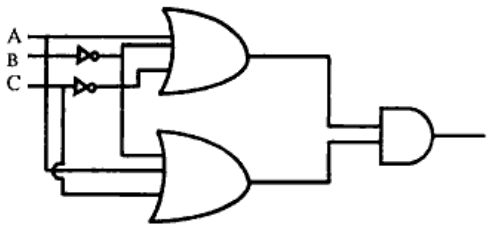
PART – A

Q.1 If $N^2 = (7601)_8$ and N is a positive integer then find the value of N.

Q.2 Find the SOP Expression of a POS Expression $F = (\overline{A} + B)(\overline{A} + C)(C + D)$.

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Q.3 Draw the reduced circuit of the given original circuit.



Q.4 What is the difference between latches and FF?

Q.5 Draw the Excitation table of all flip – flops.

Q.6 Design a NAND gate (2 input) by using 2:1 multiplexer.

Q.7 Define the following –

(i) Propagation Delay

(ii) Fan In, Fan out

Q.8 Perform the following operation –

(i) $(27)_8 + (74)_8$

(ii) $(1010)_2 - (111)_2$ by using 2's complement method.

Q.9 Explain Race – around condition in flip – flop.

Q.10 Draw NMOS inverter.

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PART – B

Q.1 Implement the minimized POS Expression for the given function -

$F(A, B, C, D) = \sum_m (4, 6, 8, 10, 11, 12, 15) + d (3, 5, 7, 9)$

Q.2 Write short note on following –

(i) Sign Magnitude Representation

(ii) Weighted and Non-weighted Codes

Q.3 Design full Adder by using 4:1 multiplexer.

Q.4 Draw CMOS NAND and CMOS NOR gate.

Q.5 What is the difference between synchronous and asynchronous counter? Design a MOD – 3 counter by using T Flip – Flop.

Q.6 (i) Design 16:1 multiplexer by using 2:1 mux only

(ii) Design 4:16 decoder by using 3:8 decoder

Q.7 What do you understand by Shift Register? What are its types? Draw their diagram along with the truth table.

PART – C

Q.1 Find the minimal sum of product for the Boolean Expression

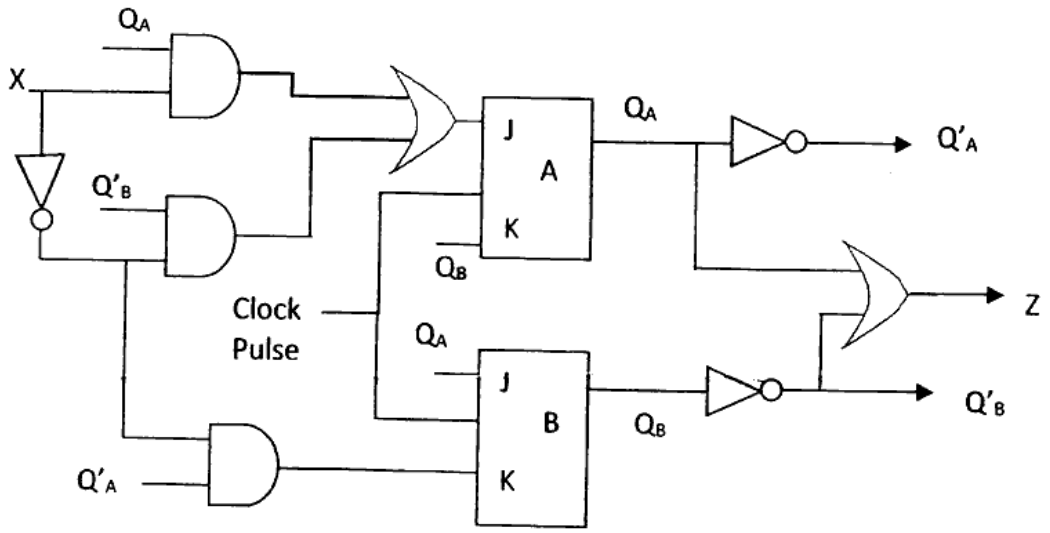
$F = \sum m (1, 2, 3, 7, 8, 9, 10, 11, 14, 15)$ using Quine – Mccluskey method.

Q.2 (i) Compare the combinational and sequential circuit.

(ii) Draw and explain TTL NAND gate win Totem pole configuration.

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Q.3 Generate a state diagram for the sequential circuit shown in figure -



Q.4 (i) Implement the following Boolean function by using 3:8 Decoder and external gates

$F(A, B, C) = \sum_m (2, 4, 5, 7)$

(ii) Design a half subtractor by using logic gates.

Q.5 (i) Perform $(68 - 34)_{10}$ in BCD.

(ii) Implement AND, EX – OR by using NOR gate.

(iii) Implement the function F by using NOR – NOR Logic -

$F = (w + \bar{x})(w + \bar{x} + z)\ (y + \bar{z})$.

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